

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Work Integrated Learning Programmes Division
First Semester 2025-2026
Mid-Semester Test
(EC-2 Regular)

Course No. : AIMLCZC418
Course Title : Introduction to Statistical Methods
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 18.01.2025 FN

No. of Pages	= 2
No. of Questions	= 6

Note to Students:

1. Please follow all the Instructions to Candidates given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Answer all the questions:

Q1. a) A sample of eleven weights (in kg) is given: 60, 72, 65, 68, 70, 100, 62, 75, 78, 80, 83.

- i) Calculate Q1, Q2, Q3, and IQR.
- ii) Determine potential outliers using the $1.5 \times \text{IQR}$ rule. [3Marks]

b) The following table shows the time (in minutes) it took for a group of astronauts to complete various training simulations: 18, 25, 19, 32, 27, 22, 29, 30, 24, 26. Find the mean and standard deviation of these times. Interpret what these statistical measures indicate about the variability and central tendency of the simulation completion times. [2Marks]

Q2 a) The length of life of an instrument produced by a machine has a normal distribution with a mean of 12 months and a standard deviation of 2 months. Find the probability that an instrument produced by this machine will last [2 Marks]

- i) less than 7 months.
- ii) between 7 and 12 months.

b) A biased coin lands heads with probability $p=0.15$. The coin is flipped $n=40$ times. [3 Marks]

- (i) Apply the binomial distribution to calculate the probability of getting exactly $k=5$ heads.
- (ii) Recall the Poisson approximation to estimate the same probability.
- (iii) Compare the two results and analyze how closely the Poisson result matches the binomial result.

Q 3. A company wants to build an automated spam email filter to classify incoming emails as either "Spam" or "Not Spam" (Ham). Given the huge volume of emails and the need for real-time performance, they choose Naïve Bayes due to its simplicity and speed. Use the Naïve Bayes classifier to predict the label of the following new email:

"Congratulations! You have won money quickly". [5 Marks]

Sample Dataset:

Email ID	Text (Bag of Words)	Label
1	Congratulations, you have won a free prize	Spam
2	Monthly meeting scheduled for Monday	Not Spam
3	Earn money quickly with this simple trick	Spam
4	Project report attached. Review by Friday	Not Spam

Q 4. A new test is developed to detect a rare genetic disorder. The disorder affects 0.1% of the population. The test has:

- Sensitivity (true positive rate): 98%

- Specificity (true negative rate): 97%

What is the probability that a person has the disorder if their test result is positive?

Q 5. Suppose a machine learning model predicts the number of errors in a document using a discrete random variable X , which takes values $\{0,1,2,3\}$. The probability mass function (PMF) of X is given as:

$$P(X = x) = \begin{cases} k & \text{if } x = 0 \\ 2k & \text{if } x = 1 \\ 3k & \text{if } x = 2 \\ 4k & \text{if } x = 3 \\ 0 & \text{otherwise} \end{cases}$$

a) Find the value of k , b) Compute Expectation of X c) What does the expected value $E(X)$ mean in this context? c) *Compute var*($2X + 3$) [5 Marks]

Q 6. A fair coin is tossed three times. The outcome of each toss is recorded as either a head (H) or a tail (T). Let X be random variable which observed the number of tails observed in

three tosses and Y be random variable represent the player's winnings(in \$), based on position of first tail according to the following rules :

“The player **wins \$3** if the **first tail** appears on the **first toss**, **\$2** if the **first tail** appears on the **second toss**, **wins \$1** if the **first tail** appears on the **third toss** and The player **loses \$2** if there is **no tail** in all three tosses”

a) Find Joint probability function of X and Y

b) Marginal Probability function for X .

c) $E(X)$

[5 Marks]
